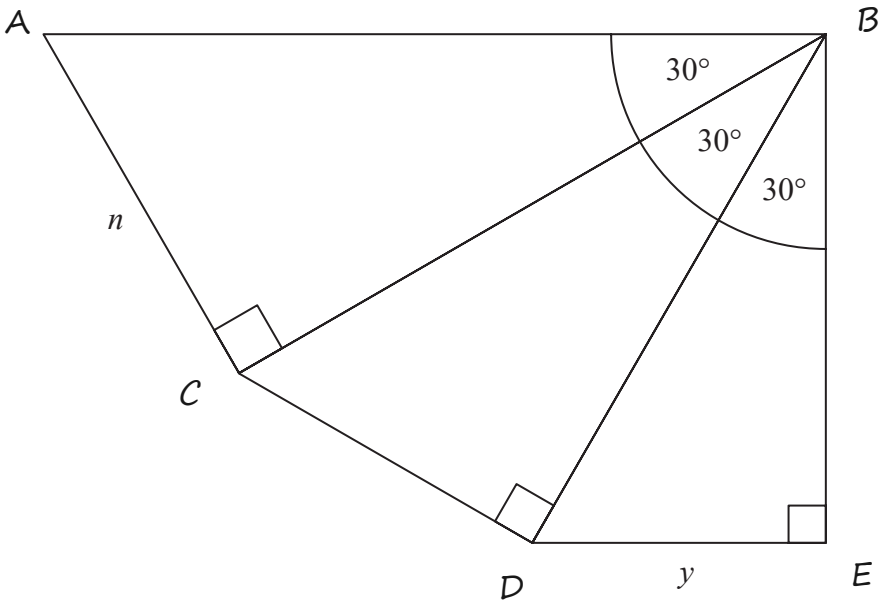
20



The diagram shows three right-angled triangles.

Prove that $y = \frac{3}{4}n$

sin30° = ½; cos30° = √3 ÷ 2.

1 mark
Exact30° values

Top triangle ABC: n ÷ AB = ½, so AB = 2n.

1 mark
Top hypotenuse2n

$BC^2 = (2n)^2 - n^2 = 3n^2$, so BC = n√3.

1 mark
Middle hypotenuse

Middle triangle BCD: BD = BC × cos30°.

BD = n√3 × (√3 ÷ 2) = 3n ÷ 2.

Bottom triangle BDE: y = BD × sin30°.

y = (3n ÷ 2) × ½ = $\frac{3}{4}$ n.

1 mark
Link all three triangles and prove y=3n/4

(Total for Question 20 is 4 marks)